THE FIRST PROPERTY OF THE ETA CONSTRUCTION: THE I-TH COHOMOLOGY OF ETA_I F IS I^I TENSORED WITH THE QUOTIENT OF H^I(F) BY ITS I-TORSION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites three cross-chapter labels (including a situation), each recorded in assembly.external_references with its canonical tag as a navigation pointer.

Editorial note. The proof cites no bibliography item inside the counted span.

Allowed standard prerequisites (editorial).

- Stalks of the eta construction. Stacks Project, Cohomology of Sheaves, Lemma lemma-eta-stalks (tag 0GT6). Reduces the isomorphism claim to a statement about the stalks, where the algebraic model applies.
- First property of the algebraic eta construction. Stacks Project, More on Algebra, Lemma more-algebra-lemma-eta-first-property (tag 0F7P). The algebraic statement that the stalkwise map is identified with.

Source reference. Original reference 0GT4: sources/stacks-chow/tags.txt, source lines 20954–20954. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0GT6: sources/stacks-chow/tags.txt, source lines 20956–20956. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0F7P: sources/stacks-chow/tags.txt, source lines 19014–19014. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *In Situation 0GT4 let $\mathcal{F}^\bullet$ be a complex of $\mathcal{I}$-torsion free $\mathcal{O}_X$-modules. There is a canonical isomorphism*

$$\mathcal{I}^{\otimes i} \otimes_{\mathcal{O}_X} \left(H^i(\mathcal{F}^\bullet)/H^i(\mathcal{F}^\bullet)[\mathcal{I}]\right) \longrightarrow H^i(\eta_{\mathcal{I}}\mathcal{F}^\bullet)$$

of cohomology sheaves.

Proof. We define a map

$$\mathcal{I}^{\otimes i} \otimes_{\mathcal{O}_X} H^i(\mathcal{F}^\bullet) \longrightarrow H^i(\eta_{\mathcal{I}}\mathcal{F}^\bullet)$$

as follows. Let g be a local section of $\mathcal{I}^{\otimes i}$ and let $\overline{s}$ be a local section of $H^i(\mathcal{F}^\bullet)$. Then $\overline{s}$ is (locally) the class of a local section s of $\mathrm{Ker}(d^i : \mathcal{F}^i \to \mathcal{F}^{i+1})$. Then we send $g \otimes \overline{s}$ to the local section gs of $(\eta_{\mathcal{I}}\mathcal{F})^i \subset \mathcal{I}^i\mathcal{F}$. Of course gs is in the kernel of d^i on $\eta_{\mathcal{I}}\mathcal{F}^\bullet$ and hence defines a local section of $H^i(\eta_{\mathcal{I}}\mathcal{F}^\bullet)$. Checking that this is well defined is without problems. We claim that this map factors through an isomorphism as given in the lemma. This we my check on stalks and hence via Lemma 0GT6 this translates into the result of More on Algebra, Lemma 0F7P. □

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